EPFL - Computer Science

COM-402 Advanced Networks

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Transporter Layer Continued

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TCP Sockets (simplified scenario)

Why both TCP and UDP?

Most applications nowadays use TCP, as this avoids re-inventing error recovery for each app. But some apps may not need TCP-like error recovery (i.e., packet retransmission):

- Voice applications / Sensor data streaming

- An application that sends just one message, like name resolution (DNS)
- Multicast (TCP does not support multicast IP addresses)

4. More TCP Bells and Whistles

How to deal with **corner cases**? Suppose that an application writes few bytes in socket; should TCP create a segment or wait for more bytes?

- Nagle's algorithm: send `min{bytes written in one RTT, MSS}`

How to handle silly window-increase advertisements by 1 byte? The idea is similar to Nagle's, we do something simple

Window field is 16 bits $\Rightarrow$ ma throughput even for RTT=1msec is $524 Mbps; can we do better$?

- negotiate a scaling factor of the window during connection setup (RFC1323)

How to detect packet loss?

Packet loss in TCP is detect via times and duplicates ACKs.

The main timers are called retransmission timeouts (RTO), it's set for every packet transmission and when a timer expires (= timeout), this is interpreted as severe loss in the channel. Thus, all timers are reset and all un-acknowledged data is marked as needing retransmission

A duplicate ACK = a TCP segment where the cumulative ACK value repeats a previously received ACK value. This is interpreted as not-so-severe loss in the channel.

Timers use an estimate of round trip time (RTT)

Why? RTO timer must be set to a value $\geq$ RTT, but not too much

What? RTT estimation gives an upper bound on the round trip time, used to set the RTO

How?